Q1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Answer:

```
ping -c 4 www.youtube.com
```

Result	Value
Destination	www.youtube.com (142.251.153.4)
Packets	4 sent, 4 received, 0% packet loss
Response times	27.8 ms, 67.5 ms, 47.9 ms, 30.0ms
Minimum / Maximum	27.780 ms / 67.533 ms
Average response time	43.305ms (about 44.6 ms)

Ping sends 4 small test packets and measures the time for each reply. All 4 packets came back, so my computer can reach YouTube. The average response time is 43.305 ms.



Output of ping to www.youtube.com

Q2. Run the Traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Answer:

```
traceroute www.flipkart.com
```

Destination: www.flipkart.com (23.212.0.112), maximum 30 hops.

Hop	Address (device)	Response time
1	192.168.207.2 (default gateway / NAT router of my Kali machine)	3.259 ms, 2.982 ms, 2.820 ms
2 to 30	* * * (no reply)	No response

Only one hop replied: hop 1, 192.168.207.2, which is the default gateway (the NAT router of my virtual machine), with about 2 to 3 ms. for hops 2 to 30 the output shows * * *, which means no reply was received. This happens because the routers after the gateway do not send back replies to the traceroute packets (they are blocked or not forwarded back through the NAT), so the other devices on the path could not be seen.



Output of traceroute to www.flipkart.com

Q3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Answer:

```
nslookup www.zomato.com
```

Item	Result
DNS server used	192.168.207.2 (port 53)
Canonical name (CNAME)	www.zomato.com is an alias of zomato.edgekey.net, which points to e70929.dsca.akamaiedge.net
IPv4 addresses of www.zomato.com	23.55.244.185 , 23.55.244.160
IPv6 addresses	2600:1417:75::687c:3673 and 2600:1417:75::687c:3613

nslookup asked the DNS server (192.168.207.2) for the address of www.zomato.com. The name points to Akamai's content delivery network, so it returns more than one IP address. The main IPv4 address is 96.17.150.105.



Output of nslookup for www.zomato.com

Q4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Answer:

Test A: My own address (to check the network setup)

```
ping -c 4 192.168.30.128
```

My Kali machine has the IP 192.168.207.130 (shown by ip addr). The ping returned 4 replies in about 0.064 ms with 0% packet loss. This only shows that the network software of my own computer works.



Output of ip addr and ping to my own address

Test B: Another device on the LAN (online)



If the device is online, the ping shows replies such as "64 bytes from ... time=... ms" and 0% packet loss.

Test C: A device that is offline (no device at this address)

```
ping -c 4 192.168.1.100
```

- **Result:** "Destination Host Unreachable" for all 4 packets, 0 received, 100% packet loss.
- **Why:** My computer asks on the LAN "who has 192.168.207.130?" (ARP), but nobody answers, so it cannot deliver the packet. The message comes from my own machine (192.168.1.128).
- **Meaning:** If a device is offline, switched off, or not on the same network, ping gets no reply and shows Unreachable or Request timed out.



Q5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: *Give a specific example for each tool based on your own test results above.*

Answer:

* **Website not loading:** Ping is the first tool to use. My ping to www.youtube.com got 4 of 4 replies with 0% packet loss, so the connection to the Internet was working; if it showed 100% loss, the problem would be in the network or the site.

* **Finding where it stops:** Traceroute shows the path. In my test to www.flipkart.com it showed only hop 1 (192.168.30.2) and then * * *, which tells me the replies stopped after my gateway.

* **Slow connection:** Ping is the most useful because it shows delay. My average time to YouTube was 44.575 ms (34.9 ms to 48.8 ms), which is normal; a high or changing time or packet loss would show a slow connection.

* **Slow connection (which part):** Traceroute shows the time at each hop, for example 4.1 to 5.0 ms at my gateway, so a hop with a large time shows where the delay starts.

* **Suspecting DNS issues:** nslookup is the best tool. It gave the IP addresses of www.zomato.com (96.17.150.105 and others) from the DNS server 192.168.30.2, so my DNS was working; an error like "server can't find" would mean a DNS problem.

* **Summary:** Ping checks if a device is reachable and how fast, traceroute finds the path and where it breaks, and nslookup checks name to IP conversion.